2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia) | 979-8-3503-5133-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/IPEMC-ECCEAsia60879.2024.10567080

AI for Smart Battery State Estimation: A Perspective

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***Abstract*-- In battery management systems (BMSs), state estimation stands as a pivotal element yet encounters significant challenges. These include the poor observability inherent in fixed configuration battery packs, limited generalizability of pre-trained machine learning models, and the deficiency of higher-level management strategies. To address these obstacles, we propose a forward-looking perspective on the future BMS state estimation, introducing the concept of a "Smart Battery". Battery digital twin enables synthetic data generation and physics-informed AI development. This approach integrates battery digital twin to generate synthetic data, which is then used for data augmentation and physics-informed AI development. Additionally, it incorporates advanced data cleaning and selection techniques to preserve essential information and augment data management efficiency. Leveraging cutting-edge AI algorithms, such as transfer learning and meta-learning, aims to mitigate issues of model generalization and feature invalidation under various operating conditions. Furthermore, this paper emphasizes the importance of multi-task learning for batteries, enabling comprehensive health assessments. By fully utilizing both short-term estimations and long-term predictions, the proposed framework contributes to the advancement of higher-level health and thermal management designs. We aim to furnish pioneering insights for state estimation in future intelligent BMSs.**

***Index Terms*—Smart Battery, Artificial Intelligence, State Estimation and Prediction, Health and Thermal Management.**

I. Introduction

Due to the high power and energy densities, high energy efficiency, and long cycle lives, lithium-ion batteries and the associated battery management systems (BMSs) have emerged as an indispensable technology in the sustainable shift of energy storage and electric mobility. BMSs focus on three key aspects: the basis, strategy, and target of management, as summarized in Fig. 1 [1]. By leveraging key operational parameters, i.e., voltage, current, temperature, and time, effective management strategies can be designed, which include security management to ensure operation within safe parameters, equilibrium control for balancing charge and discharge cycles, thermal management to regulate temperature, and an information management system for data integrity and accessibility. These strategies collectively work towards ensuring the battery operates safely, maintaining balance, and performing efficiently. Central to realizing these objectives is the precision in battery state estimation.

The rapid advancement of artificial intelligence (AI) has revolutionized the state estimation processes BMSs for lithium-ion batteries, offering more robust and precise assessments of battery performance and health [2], [3]. Key focus areas include state of charge (SOC) estimation [4], state of temperature (SOT) estimation [5], state of power (SOP) estimation, state of health (SOH) estimation [6], aging knee point detection, and lifetime prediction [7], which are increasingly critical in optimizing the performance and reliability of battery systems. Despite the strides in feature extraction and machine learning (ML) performance enhancement, state estimation within BMSs remains facing several challenges, as listed below.

Poor model interpretability: Until now, nearly all the AI models in battery management are "black box" in nature. These AI models lack interpretability, struggling in unfamiliar scenarios due to non-linear characteristics of batteries. Achieving accurate and interpretable predictions for long-term health and performance under diverse conditions is a major research gap.

Poor observability of battery pack with fixed configuration: In fixed configuration battery packs, cells are connected in series or parallel in a predetermined way. This rigid connection limits the scope of obtainable data to cell- and pack-level voltage, current, and few numbers of temperature readings. Consequently, state estimators are constrained to be designed with this limited dataset, which can inhibit a comprehensive understanding of the battery's actual condition.

Limited generalizability of pre-trained ML model: For traditional ML, the objective is to achieve optimal performance on tasks that the model has been trained on. However, the pre-trained ML models might not perform well on new tasks, especially those under different working conditions from the training data. This challenge arises due to the variability in data and these models typically require retraining, which can be time-consuming and data intensive.

Insufficient states information: Traditional BMSs only obtain limited information about battery internal states such as SOC, SOH. Despite studies of obtaining other states such as SOT and SOP, how to obtain comprehensive state information in a systematic way is still unclear. This brings challenges to intelligent control and optimization that requires more information about battery internal states.

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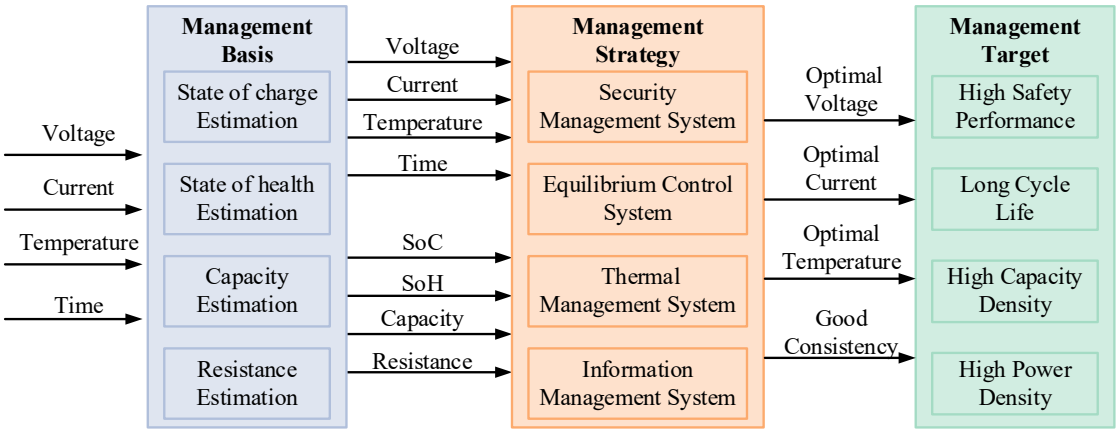

Fig. 1. The functionality of traditional BMSs.

Underutilization of estimated states: Present BMSs are adept at operational monitoring but often lack the ability to effectively use these detailed insights for broader system optimization and management. This deficiency reveals a need for more strategic applications of the obtained performance and health states, which would enhance the overall efficiency, reliability, and longevity of battery systems.

Recognizing these challenges, this paper introduces an innovative approach for the next generation of BMS state estimation using AI, conceptualizing the "Smart Battery" framework [10].

II. AI-BASED STATE ESTIMATION AND MANAGEMENT IN SMART BATTERY

Smart Battery, as an intelligent BMS, aims to optimize performance and extend their usable lifetime by closely monitoring several key states, as shown in Fig. 2. The SOC, representing the current charge level relative to the battery's capacity, is vital, as it quantifies the remaining energy in the battery. SOT is monitored to safeguard against the thermal extremes that can significantly affect battery efficiency and safety. Meanwhile, SOH offers insights into the degradation and overall condition of the battery, which informs predictions about its lifetime. Additionally, understanding the knee point, i.e., the stage at which the battery's capacity starts to decline more rapidly, is essential for predicting the transition from optimal performance to reduced efficiency. By closely tracking these states, Smart Battery can adjust bypass strategies in real-time, ensuring balanced performance and consistent aging rates across the battery cells. This approach not only enhances the overall reliability but also aligns the aging trajectory of individual cells, thereby optimizing the battery pack's efficiency. Furthermore, Smart Battery can provide advance warning when the batteries are approaching accelerated aging or the end of cycle life. Such proactive notifications enable timely maintenance interventions or the scheduling of replacements, thus aiding in timely maintenance and replacement decisions. As an integral component of Smart Battery, AI technology plays a pivotal role, with its capacity to process and analyze complex, non-linear relationships within the collected data. AI algorithms can learn from historical data, predict future aging or thermal behavior, and make decisions to optimize battery operation.

A. *Battery Digital Twin*

Battery digital twin serves as highly accurate virtual models of physical battery systems, capturing their behaviors in real-time [8]. It incorporates comprehensive physical modeling such as electrochemical model or pseudo-two-dimensional model, specific on-field data collection, and simulation environments tailored to the system of interest, enabling precise replication of the physical counterpart's behavior across various conditions and time scales. Physical model tools such as PyBaMM, COMSOL, Ansys, and Batemo can be used.

One aim of a digital twin is to simulate various conditions and time scales, closely mirroring the real-world behavior of its physical counterpart. Therefore, generation of vast amounts of synthetic data is enabled. For example, Ruan et al. propose a deep learning health diagnostic framework that leverages synthetic data generated from full-cell and half-cell open circuit voltages of fresh batteries, avoiding the need for extensive experimental aging data [9].

Moreover, the research demonstrates a shift towards physics-informed AI, beyond purely data-driven AI, by incorporating physical insights into the learning process, thus ensuring that predictions not only fit the observed data but also adhere to known physical behaviors. This approach not only enables the creation of synthetic data for training and testing models but also enhances the understanding of physical phenomena affecting battery health.

B. *Data cleaning and selection*

Due to the increased availability of public battery datasets, the majority of the models developed in the literature are both trained and validated on lab-generated data [2]. Using data generated in closely controlled conditions such as current and temperature to build models can capture the performance of the battery under these conditions. However, the ultimate goal of smart batteries

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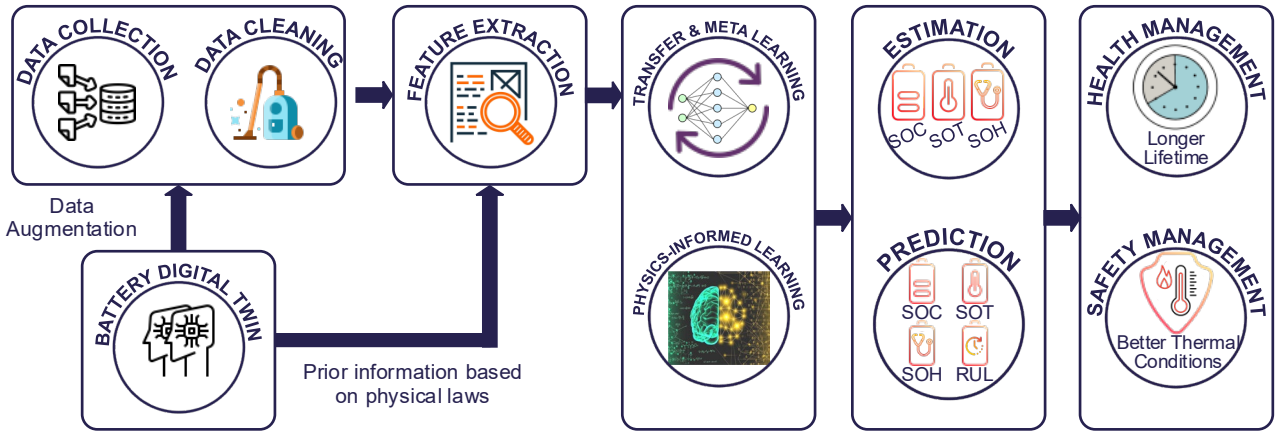

Fig. 2. State estimation in Smart Battery.

is to use the derived models in a real life setting for online estimation and prediction. It is thus paramount that models designed for use in BMS operation are also robust for in-field data which is more representative of the end use case. In-field data is generally more difficult to work with. Sensor noise, missing or false data, large current variations, uncontrolled temperature conditions and a small range of operation conditions makes it challenging to extract features and ensure a high performance of the models.

To combat the messiness of in-field data, data cleaning methods can be effectively implemented to enhance the quality of the data and thus improve the performance of the models. Previous studies have shown that ML can be a key solution to these problems. This includes the effective integration of data imputation technique [11], filling in missing data values [12], correcting outliers, and denoising [13].

Another emerging area in online battery estimation is the use of data selection techniques to strategically select high quality data segments to update the estimated parameters. While the aim of this method is not to restore missing data or remove sensor noise, the goal is still to only feed data of the highest quality to the models. By leveraging parameter sensitivity to measurement data, significant improvement can be made by only considering data segments where this sensitivity is highest, as recently shown by [14].

Additionally, unlike traditional approaches that rely on fixed input datasets, Smart Battery capitalizes on random and partial charging curves. This is made possible by incorporating an array of switches within the battery packs, facilitating the design of novel software-based sensors and actuators. This redesign of BMS architecture unlocks the potential for more dynamic BMS functionalities. By employing actions such as bypassing, more information can be obtained—including open-circuit voltage, voltage response during relaxation periods, online operando electrochemical impedance acquisition, and internal resistance calculation. Moreover, data on temperature gradients and voltage inconsistencies become accessible, providing insight for accurate battery health assessment.

C. *Advanced AI algorithms*

The task of health assessment for batteries includes multi-time scale estimation of battery, e.g., SOH estimation, degradation trajectory and aging knee identification, and lifetime prediction [15]. Traditionally, AI-based health assessment methods have been streamlined into sequential stages: input preparation, model training, and subsequent evaluation. To ensure the effectiveness of models deployed in real-time BMS operation, in-field data needs to be included in the model design process to help bridge the gap between the lab and the BMS. Data cleaning and data selection can be useful tools for this, but a major limitation of using in-field data in research is its availability. Therefore, Smart Battery is

TABLE I

COMPARISON OF THREE DISTINCT PARADIGMS OF AI

Methods	Machine learning	Transfer learning	Meta learning
Objective	▪ Optimal performance on trained tasks	▪ Knowledge transfer to new tasks	▪ Rapid task adaptation with few examples
Support Set	▪ N/A	▪ Used to guide fine-tuning process	▪ Crucial for defining class prototypes
Query Set	▪ N/A	▪ Used to test the adaptability of the transferred knowledge	▪ Used to adjust model during training to new examples
Evaluation	▪ Performance assessed on a fixed test set	▪ Performance assessed on a new task after fine-tuning	▪ Performance assessed on new tasks with few examples

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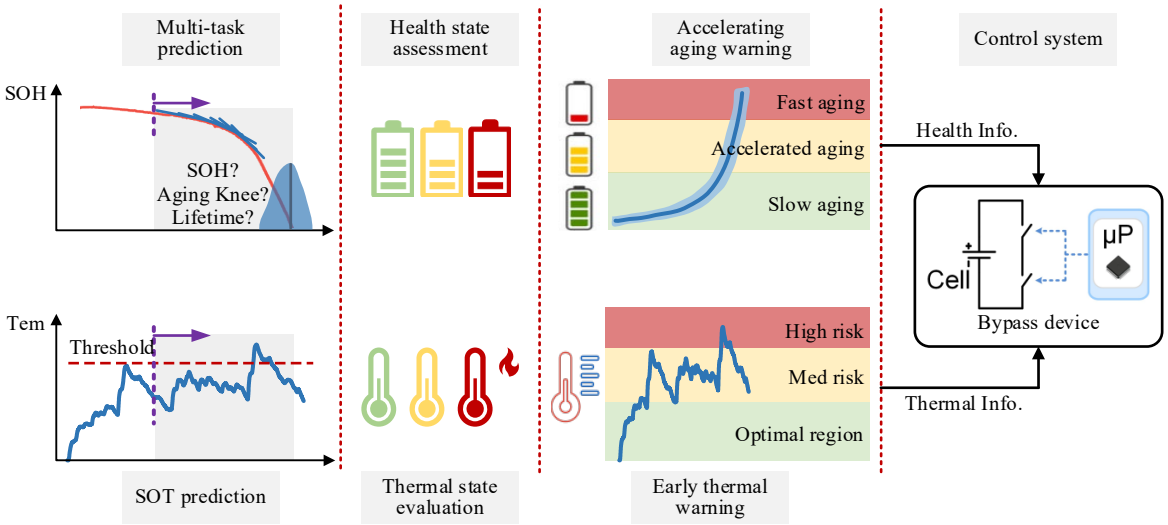

Fig. 3. Flowchart for health and thermal management in Smart Battery.

poised to substantially advance the field of BMS and health assessment by accommodating randomness in operation, adopting transfer and meta-learning techniques, and refining multi-time scale estimations.

There are three distinct paradigms that are shaping AI: conventional ML, transfer learning, and meta learning, where the comparison is summarized in Table I. Traditional ML models, though effective in certain scenarios, are limited by their dependency on extensive, labeled datasets and suffer from inflexibility when faced with new tasks, often requiring comprehensive retraining. For further enhancing the health assessment, transfer learning and meta-learning will be developed to replace conventional ML. Transfer learning focuses on utilizing knowledge acquired from one task to enhance performance on a different, yet related task. It leverages pre-trained models and fine-tunes them with minimal parameter adjustments to adeptly tackle new and related tasks. This not only speeds up the adaptation process but also saves computational resources. Meta-learning pushes the boundaries even further by enabling rapid adaptation to novel tasks with very few examples. By simulating real-world scenarios through episodic training and adopting a meta-strategy that focuses on the acquisition of knowledge from limited data points, meta-learning surpasses traditional methods.

Additional, Smart Battery systems enhance multi-time scale state estimations using multi-task learning [7], extending from second-scale SOC to minute-scale SOT, month-scale SOH, and long-term degradation trajectories, to year-scale aging knee points and overall cycle lifetime.

D. Battery health and thermal management

Despite progress in algorithms development for enhancing the accuracy of battery states estimation, a significant gap remains in the effective utilization of these estimates in health and thermal management. Health and safety management is essential to ensure the battery works under safe conditions to reduce customer anxiety and the risk of thermal runaway, and in the meantime maximize the lifetime of batteries. Although health and thermal status are predicted based on advanced algorithms, the upper management is still ignored by the existing BMSs. Therefore, efficient health and safety management is supposed to be designed in future BMSs where bypass actions will be used in the smart battery pack to manage batteries in an optimal way.

A potential framework, as shown in Fig. 3, outlines a smart control system for managing batteries. This system uses prognostics and predictions about the battery's future health and thermal states. The battery states are categorized into three stages: "green" for slow aging or optimal thermal conditions, "yellow" for accelerated aging or medium thermal risk, and "red" for rapid aging or high thermal risk. Comprehensive health predictions offer insights into the battery's aging process, guiding timely management strategies that can prolong battery life. For instance, balancing and bypassing will be triggered effectively during accelerated aging stages to slow down the aging rate. Simultaneously, the system monitors the battery's thermal behavior in real-time and predicts short-term temperature changes. Based on these predictions, the control strategies adjust to keep the batteries within safe thermal limits. By combining long-term health management with real-time thermal monitoring, the system ensures batteries operate safely and efficiently, maximizing power usage and extending their lifetime.

III. Conclusion

By introducing the "Smart Battery" framework, future prospects on the use of AI methods to enhance state estimation in intelligent BMS is systematically proposed. The exploration of the state estimation framework encompasses four key perspectives, i.e., high-fidelity battery digital twin, effective data processing, advanced model training, and the practical application of these estimations in health and thermal management. This

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approach not only enhances data quality and model adaptability but also emphasizes the strategic use of estimated states for optimizing battery performance. Smart Battery with its incorporation of cutting-edge AI algorithms such as physics-informed AI, transfer learning, meta-learning, and multi-task learning, marks a significant step forward in the field of battery management. It offers a novel insight for extending battery life and ensuring safe and efficient operation.

ACKNOWLEDGMENT

This work was funded by the "SMART BATTERY" project, granted by Villum Foundation in 2021 (project number 222860).

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